

Latchan Chhetri

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[Halo Mind Research Group Portfolio](#) | [LinkedIn](#) | [GitHub](#)

Profile & Research Focus

AI & Machine Learning Researcher and Architect specializing in Computer Vision, Sub-Quadratic Sequence Modeling (Mamba/SSMs), Bio-Med AI, Representation Learning, and High-Throughput Deep Learning Architectures (Edge Frameworks). Founder and Chief AI Researcher of Halo Mind Research Group, with lead-author publications and preprints under peer review across IEEE/ ACM/ Springer/ T&F venues. Proven track record in designing custom gating mechanisms, stabilizing representation drift, and engineering modular pipelines for complex sequential and spatial data. Focused on building hardware-efficient, scalable neural architectures for frontier AI systems.

Education

Sikkim Manipal Institute of Technology (SMIT)

Bachelor of Technology in Computer Science and Engineering (AI & Data Science)

Sikkim, India

July 2023 – June 2027

St. Xavier's School

ISC (Class 11 & 12) – Science

West Bengal, India

April 2021 – March 2023

St. Xavier's School

ICSE (Class 1 – 10)

West Bengal, India

April 2010 – March 2021

Research Manuscripts

Peer-Reviewed & Accepted

- A. Kumar*, L. Chhetri*, D. Das* (*Equal Contribution). "SPECTRAFORGE: Domain-Equalized Frequency-Spatial Fusion for Synthetic Dermatology Detection." *Accepted to the IEEE International Conference on Data Science and Advanced Analytics (DSAA), 2026. [CORE A Tier | <25% Acceptance Rate]*
- L. Chhetri, A. Kumar. "DDV-GNet: High-Throughput Defect Detection for Space Manufacturing via Deep Delta Gated Networks." *Accepted for Oral Presentation at IEEE SPACE, 2026.*
- L. Chhetri, A. Datta, P. Ghosal. "Attention-Enhanced Swin Transformers for Robust Brain Tumor Classification Under Patient-Level Data Splitting." *Accepted at IEEE GCON, 2026.*
- L. Chhetri, A. Kumar. "Optimizing Deep Learning for Brain Tumor Classification: A Comparative Ablation Study of Preprocessing and Augmentation Strategies." *Accepted at IEEE GCON, 2026.*
- A. Mukherjee, A. Kumar, S. Bala, SR. Varma, A. Anand, H. Das, L. Chhetri. "Risk-Controlled Urban Change Detection: Conformal Prediction Wrappers for Provable Reliability in High-Resolution Satellite Imagery." *Accepted for Oral Presentation at ICCI, 2026.*
- R. Das, L. Chhetri, A. Kumar, P. Ghosal. "Intrinsic Neural Firewalls for Cyber-Physical Systems: Robust Anomaly Rejection via Deep Delta Residual Overwrites." *Accepted for Oral Presentation at WIN 6.0, 2026.*

Under Review

- L. Chhetri, A. Kumar, P. Ghosal. "DeltaGate-CNN: A Lightweight Convolutional Architecture for Efficient Remote Sensing Image Classification" *Under Review at IEEE INDICON, 2026.*
- L. Chhetri, A. Kumar, D. Das, P. Ghosal. "Interpretable Solar Panel Defect Detection via Fuzzy Rule Extraction from Hierarchical Vision Models." *Under Review at ICVGIP, 2026.*
- A. Kumar, L. Chhetri, A. Anand, H. Das, A. Mukherjee. "Beyond Limited Labels: Safe Semi-Supervised Learning for Malaria Diagnosis." *Under Review at ICVGIP, 2026.*

Experience

Founder & Chief AI Researcher

Halo Mind Research Group

Jan 2026 – Present

Independent Research Organization

- Architectural Direction: Direct the fundamental scientific trajectory of an independent research group focused on hardware-efficient networks & sub-quadratic mechanisms.
- Research Leadership: Manage a distributed team of ML Systems Engineers, Applied Vision Researchers, and Data Engineering Interns to execute end-to-end experimental designs.
- Open-Source Development: Oversee the deployment and rigorous evaluation of robust PyTorch frameworks for space AI, solar-weather forecasting, medical imaging, and conformal prediction's.

Research Assistant (Satellite AI & Causal Inference)

AI & Global Development Lab (AIDevLab) – UT Austin / Chalmers

Jun 2026 – Present

Remote

- Spatial-Temporal Conformal Frameworks: Collaborating under the supervision of Prof. Connor T. Jerzak and Prof. Adel Daoud to engineer spatial-temporal conformal inference frameworks for satellite-imagery based poverty prediction models, demonstrating how standard conformal intervals break down under large geographic shifts.
- Model Evaluations: Engineered custom Data-Loaders for continuous International Wealth Index (IWI) score prediction, establishing baseline prediction errors and conformal coverage metrics across geographic hold-outs (e.g., training on African data, evaluating on out-of-continent hold-outs).

Research Intern - (Computer Vision, OCR & Document Analytics)
Indian Statistical Institute, Kolkata

Jun 2026 – Present
Hybrid

- Geometric Recovery Architecture: Spearhead research under Prof. Umapada Pal to replace Thin-Plate Splines (TPS) with a sub-quadratic Vision Mamba (SSM) backbone, predicting dense 2D displacement fields to unwarp discontinuous metallic folds.
- Forensic-Safe Semantic Recognition: Engineered a 3-stage pipeline integrating a Vision-Language Model (Qwen-VL) with a novel Conformal Risk-Controlled Abstention module, calculating normalized confidence scores to eliminate LLM hallucinations on destroyed texts.

Research Intern - (Computer Vision, OCR & Document Analytics)
The University of Salford, UK

Jun 2026 – Present
Remote

- Synthetic Dataset Engineering: Developed a scalable, automated dataset generation pipeline via Blender’s Python API (bpy) under Prof. Shivakumara Palaihaiahnakote, simulating extreme physical metal buckling and volumetric mud occlusion.
- Mathematical Ground-Truth Extraction: Programmed the rendering pipeline to export uncompressed 32-bit OpenEXR multi-layer files, preserving pure floating-point vector coordinates for strict geometric regression.
- Domain Generalization & Benchmarking: Curating a real-world physical “Crash-ALPR” test set to benchmark zero-shot domain adaptation against SOTA deformable rectifiers (e.g., ABINet, MORN), mathematically isolating geometric failure modes.

Research Assistant & Team Lead
Sikkim Manipal Institute of Technology (SMIT)

July 2025 – July 2026
Onsite

- Medical Vision & Attention Mechanisms: Architected attention-enhanced Swin Transformer pipelines for complex medical diagnostics. Delivered robust brain tumor classification capabilities and generalized feature extraction under rigorous patient-level data splitting protocols (Accepted, IEEE GCON).
- Fuzzy Logic & Medical Image Segmentation: Engineered *AHF-RBF Net* by replacing traditional Gaussian fuzzy membership with learnable Radial Basis Function (RBF) kernels. This architecture provides numerically stable, boundary-aware spatial attention for complex lesion segmentation, achieving a superior 84.55% IoU on the ISIC 2016 benchmark.
- Cyber-Physical Security & Anomaly Rejection: Engineered *Intrinsic Neural Firewalls* utilizing Deep Delta Residual Overwrites for edge-deployed cyber-physical systems. Achieved high-performance, zero-shot anomaly rejection against False Data Injection Attacks (Accepted for **Oral Presentation**, WIN 6.0).
- Research Leadership & Pipeline Engineering: Directed and mentored student research teams, training junior researchers in core deep learning methodologies and guiding end-to-end experimental design from conceptualization to multiple first author and co-authored acceptances in IEEE and Springer venues.

AI & Data Science Intern
Soft Nexis Technology

Jul 2025 – Aug 2025
Remote

- Predictive Analytics Pipeline: Engineered and deployed **RealVisor**, an end-to-end AI real estate platform. Developed robust predictive pipelines utilizing XGBoost and Random Forest Regressors to estimate property valuations based on complex spatial features.
- Model Evaluation & Deployment: Conducted rigorous evaluation across curated real-world datasets, strictly outperforming baseline models, and designed a production-ready Streamlit dashboard featuring interactive market trend visualizations and automated investment analysis.

Patents & Intellectual Property

L. Chhetri, et al. “*Halo-Hydra: A Physics-Informed Generative Framework for Multimodal Visual Restoration in Underwater-Turbid and Degraded Scattering Environments.*” Intellectual Property India / PCT, Patent Filed / Pending, 2026.

- Engineered a physics-guided generative architecture to perform real-time visual enhancement and denoising across highly degraded, non-linear scattering media on underwater imagery.
- Designed for embedded and edge-level deployment, optimizing parameter efficiency while preserving structural spatial integrity without relying on clean reference ground truths.

Active Research & Projects

Degraded Ancient Ashokan Brahmi Script Recognition (ML/CV Lead)
Targeting ICDAR (2027) | PyTorch, WGAN-GP, SimCLR, OCR Pipelines

- Leading a computer vision team alongside A. Anand, A. Kumar, and G. Sarma to engineer a data generation and ETL pipeline using WGAN-GP to synthesize 20K+ Brahmi characters, applying physically-motivated degradation to construct a 150K sequence dataset.
- Developed an end-to-end OCR architecture integrating SimCLR self-supervised pretraining on a ResNet34 backbone with a BiLSTM-CTC decoder.
- Conducted a synthetic-to-real domain gap study, establishing the first severity-based Character Error Rate (CER) evaluation benchmark for ancient Indic scripts.

DDL-Mamba – Geometric Residual Learning for 3D Medical Image Segmentation
Under Review at Pattern Recognition Journal (Q1) | PyTorch, U-Mamba, DDL, 3D Medical Imaging

- Developing a novel **Delta-Mamba** framework integrating **Deep Delta Learning (DDL)** blocks into a U-Mamba backbone for 3D volumetric medical image segmentation, targeting concentric cardiac structures on the **ACDC** and **Synapse** benchmarks.
- Engineered a **DDL** block with **L2-normalized, Softplus-constrained** weights to enforce strictly non-negative global slot amplitudes, stabilizing the geometric manifold across deep encoder layers and resolving the representation drift inherent in dense 3D State Space Models.

- Implemented β -gate visualization providing a geometrically interpretable, layer-wise explainability mechanism for segmentation decisions — structurally impossible in standard Transformer or prior Mamba-based architectures.

Ionospheric TEC Forecasting

Under Review at Advances in Space Research (SCIE Q2) | PyTorch, Deep Delta Learning, Conformal Prediction

- Co-authored a CNN-DDL architecture featuring a dynamic β -gate that conditions real-time solar wind drivers to scale non-linear corrections over physical persistence baselines.
- Evaluated via zero-shot cross-solar-cycle testing (SC25 to SC24), achieving SOTA storm-time error rates (2.30 RMSE) and outperforming deep sequential baselines and IRI-2020.
- Integrated Marginal Split Conformal Prediction to establish distribution-free uncertainty quantification, providing a mathematically reliable 90% empirical coverage bound during extreme geomagnetic superstorms.

AHF-RBFNet – Medical Image Segmentation

Targeting Expert Systems with Applications (Q1) | PyTorch, AHF-Net, Fuzzy Logic

- Collaborating under the supervision of Dr. Swalpa Kumar Roy and Prof. Palash Ghoshal to engineer AHF-RBFNet by integrating a learnable Radial Basis Function (RBF) fuzzy membership kernel into an Attention-Guided Hierarchical Fusion U-Net to dynamically filter semantic uncertainty.
- Achieved SOTA performance on the ISIC 2016 dataset (84.55% IoU, 91.97% Dice), strictly outperforming 11 baselines including the original AHF-U-Net architecture.
- Orchestrated massive hyperparameter optimization via Optuna and executed rigorous multi-dataset evaluations (ISIC, BUSI, PH2).

Honors, Certifications & Academic Service

- **Technical Peer Reviewer:** Invited and served as an official peer reviewer for the **IEEE GCON** conference, evaluating manuscripts in applied deep learning and computer vision.
- **Top 1% Topper & Gold Medalist:** Introduction to Internet of Things, NPTEL (Funded by MoE, Govt. of India). Elite score: 91%.
- **Deep Learning & AI Certifications:** NLP with DL, PyTorch for Medical Imaging (Udemy); GenAI for Data Scientists (Coursera); Applied Machine Learning (AMII).

Technical Skills

Languages & Frameworks: Python, C, Java, PyTorch, TensorFlow, Keras, Blender, Streamlit, LaTeX, Git, GitHub

Computer Vision: Vision Transformers, Mamba, SSMS, U-Net, Deep Delta Learning, Diffusion Models, GANs, YOLO, CNN's

Signal Processing & Time Series: 1D CNNs, Signal Analysis, Spacy, NLTK, Text Mining, OCR Pipelines

Data Engineering & Tools: Pandas, NumPy, SciPy, Scikit-learn, ETL Workflows, Hadoop, Spark, Computational Workflows, High-Throughput Image Processing